

Course Code: CS413
Full Marks=50

Course Title: Database Management Systems

Time=2 hours

Instructions: There are a total of 6 questions. You need to attempt any one question Q5 or Q6. All other questions are compulsory. Missing data, if any, may be suitably assumed.

- 1(a).** Consider a relation $R(P, Q, R, S, T, U, V)$ and its decomposition into 5 sub-relations:
 $R_1(P, Q, R)$, $R_2(Q, S)$, $R_3(R, T)$, $R_4(S, U)$ and $R_5(U, V)$
Suppose the following functional dependencies hold on R :
 $P \rightarrow Q$, $Q \rightarrow S$, $R \rightarrow ST$, $S \rightarrow U$, $U \rightarrow V$
Determine whether the decomposition of R is lossless or lossy. Show all intermediate steps. [5]
- 1(b).** Determine whether each decomposition as mentioned in Q1(a) has the dependency preservation property or not. Show all intermediate steps. [5]
- 2.** Consider the relation $R(A, B, C, D, E, F, G, H, I, J)$ with the following functional dependencies:
 $A \rightarrow BC$
 $C \rightarrow D$
 $D \rightarrow EI$
 $B \rightarrow F$
 $A \rightarrow GJ$
- a) Identify all candidate key(s) for relation R . Identify the prime and non-prime attributes. [2]
 - b) Normalize the relation to Second Normal Form (2NF). Show all decompositions and explain your reasoning. [1]
 - c) Further normalize to Third Normal Form (3NF). Indicate if decomposition is needed and provide resulting relations. [2]
 - d) Finally, normalize the relation(s) to Boyce-Codd Normal Form (BCNF). Justify your answer. [3]
 - e) How many possible super keys are present? [2]
- 3(a).** A relation $R(A, B, C, D)$ having two FD sets F_1 and F_2 :
 $F_1 = \{A \rightarrow B, B \rightarrow C, AB \rightarrow D\}$ and
 $F_2 = \{A \rightarrow B, B \rightarrow C, A \rightarrow C, A \rightarrow D\}$.
Check whether F_1 and F_2 are equivalent or not. Show all intermediate steps with justification. [5]
- 3(b).** The following functional dependencies hold true for the relational scheme $R(A, B, C, D)$
 $A \rightarrow B$, $B \rightarrow C$, $D \rightarrow ABC$, $AC \rightarrow D$
Write the irreducible equivalent (minimal cover) for this set of functional dependencies. [5]
- 4.** Consider the following **University Database** schema, where the primary keys are underlined.
Students(StudentID, Name, Major, Year)
Courses(CourseID, CourseName, Department)
Enrollments(EnrollmentID, StudentID, CourseID, Grade)
Instructors(InstructorID, Name, Department)
- Write the following queries in SQL considering the above Insurance relational schema.
- a) Write a query to list the names of students who are enrolled in any course offered by the 'Computer Science' department. [1]
 - b) Update the major of a student with StudentID = 101 to "Data Science". [1]
 - c) Find the courses that have more than 5 students enrolled, and list them with the number of students. [2]

- d) For each major, find the average grade of students and list only those majors where the average grade is greater than 75. Display results in descending order of average grade. [3]
- e) Find the names of students who are enrolled in at least one course where their grade is greater than 80. [3]

5(a) Consider the following schedule S of transactions T1, T2, T3, T4: [7]

T1	T2	T3	T4
	R(a)		
		W(a)	
	W(a) Commit		
		R(c) Commit	
W(b) R(c) Commit			
			R(a) W(b) Commit

- Check whether the given schedule S is conflict serializable or not. If yes, then determine all the possible serialized schedules.
- Check whether the given schedule S is Recoverable schedule or not?
- Check whether the given schedule S is cascadeless schedule or not?

5(b). Consider a Table T with a key field K. A B-Tree of order P where P denotes the maximum number of record pointers in a B-Tree. Assume K is 16 bytes long, disk block size is 2046 Bytes, record pointer is 8 bytes and block pointer size is 4 bytes long. What is maximum value of P so that each B-Tree node can completely fit in a single disk block. [3]

6(a). Check whether the given schedule S is view serializable or not. If yes, then give the serial schedule. [5]

S: R₁(B), W₂(R), W₃(B), W₁(B), R₃(B), W₂(B)

6(b). Find the minimum number of tables required to represent the given ER diagram (Fig-1) in relational model. Show all required tables along with their corresponding attributes and keys. [5]

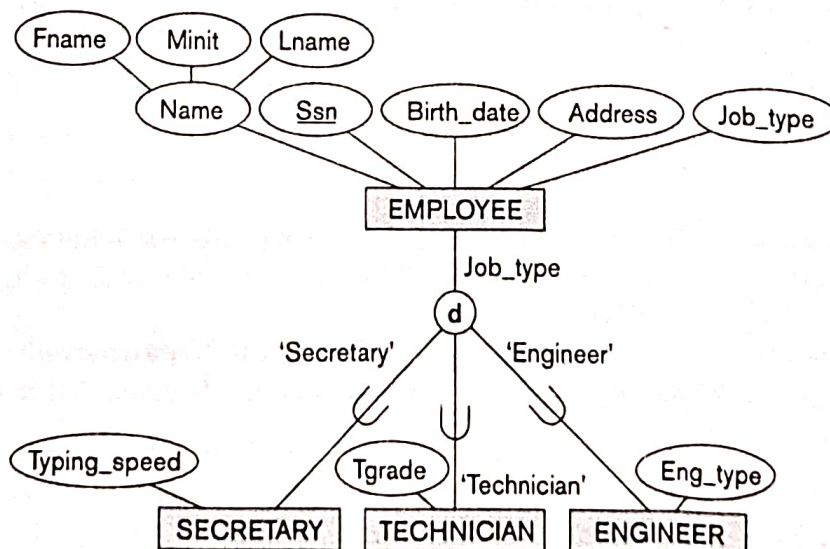


Fig-1